

Overfitting and Regularization - Study Notes

Overfitting and underfitting

Overfitting happens when a model learns the noise in the training data. It shows very high training accuracy but poor accuracy on validation and test data. Underfitting happens when the model is too simple to capture the pattern, so it performs poorly on both the training and the validation data.

Bias-variance tradeoff

High bias means the model is too simple and underfits. High variance means the model is too sensitive to the training data and overfits. The goal is a balance between the two. Learning curves help: a large gap between training error and validation error signals overfitting.

Ways to reduce overfitting

Get more training data, use a simpler model, remove irrelevant features, apply regularization, use dropout in neural networks, use early stopping, augment the data, and evaluate with cross-validation.

L1 and L2 regularization

Regularization adds a penalty on the size of the model weights to the loss function. L2 regularization, also called Ridge, penalizes the sum of the squared weights. It shrinks weights toward zero but rarely makes them exactly zero. L1 regularization, also called Lasso, penalizes the sum of the absolute values of the weights. It can push some weights to exactly zero, which produces sparse models and works as automatic feature selection. Elastic Net combines both penalties. The regularization strength is controlled by a hyperparameter, often called lambda or alpha: a larger value gives a simpler model.

Dropout

Dropout is a regularization technique for neural networks. During each training step it randomly sets a fraction of the neuron outputs to zero, which prevents neurons from relying on each other too much. Typical dropout rates are between 0.2 and 0.5. Dropout is switched off at inference time.

Early stopping

Early stopping monitors the validation loss during training and stops when it has not improved for a number of epochs, called the patience. The weights from the best epoch are then restored.